CS690A: Assignment 1

Comparing Foundation Models against a PCA-based Pipeline for Single-Cell Clustering

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1 Overview

This report describes the clustering pipelines built for Task 1.1 (PCA-based Leiden clustering), Task 1.2 (foundation-model-based clustering using scGPT and Geneformer), and Task 2 (a combined pipeline using PCA and scGPT embeddings). All pipelines were evaluated on the same single-cell RNA-seq dataset (10,727 cells, 12,303 genes) provided for the assignment, with clustering quality assessed via Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) against held-out ground truth on Kaggle, alongside label-free silhouette scores.

2 Task 1.1: PCA-based Leiden Clustering

2.1 Pipeline

1. **Quality control**: Cells retained with 900–4000 detected genes (10,482 / 10,727 cells passed, 97.7%).
2. **Feature selection**: Top 2,000 highly variable genes (Seurat flavor).
3. **Dimensionality reduction**: Scaling (zero-centered, clipped at max value 10) followed by PCA to 50 components, capturing 26.3% of total variance.
4. **Clustering**: Neighborhood graph (15 neighbors, 50 PCs) followed by Leiden clustering, with resolution swept over $\{0.10, 0.15, \ldots, 2.00\}$ and selected by silhouette score.
5. **Back-filling**: The 245 cells removed by QC were assigned labels via a k-nearest-neighbor classifier (k=15) trained on the QC-passed cells' PCA coordinates, so that the final submission covers all 10,727 cells.

2.2 Resolution Selection

The resolution sweep is summarized in Table 1. Resolution 0.15 was selected as the best trade-off, yielding 12 clusters with silhouette 0.2495.

2.3 Cluster Annotation

Marker genes for each of the 12 clusters were identified via a Wilcoxon rank-sum test (`scanpy.tl.rank_genes_group`
Based on canonical PBMC/bone-marrow marker sets, clusters were annotated as shown in Table 2.

This annotation is consistent with a typical PBMC/bone-marrow composition, suggesting the clustering captures genuine biological structure rather than purely technical variation. Note: Due to Leiden's stochastic elements, the exact cluster count can vary by ±1 between runs at the same

Table 1: PCA + Leiden resolution sweep (selected row in bold)

Resolution	# Clusters	Silhouette
0.10	11	0.2442
0.15	**12**	**0.2495**
0.20	14	0.2294
0.25	15	0.1787
0.30	15	0.1797
0.50	20	0.1910
1.00	22	0.1779

Table 2: Cluster annotations based on top marker genes

Cluster	Cell Type	Key Markers
0	Naive T cells	LDHB, RPS29, RPS15A, TPT1 (ribosomal-high)
1	Plasmacytoid Dendritic Cells (pDCs)	PLD4, CD74, HLA-DPA1, GSN
2	Classical Monocytes (CD14+)	LYZ, S100A9, S100A8, FCN1
3	Natural Killer (NK) cells	GNLY, NKG7, PRF1, KLRD1
4	B cells	CD79A, MS4A1, CD79B, BANK1
5	CD8+ Memory/Effector T cells	CCL5, NKG7, GZMA, GZMK
6	Non-classical Monocytes (CD16+)	LST1, AIF1, FCER1G, MS4A7
7	Platelets / Megakaryocytes	PF4, PPBP, NRGN, GNG11
8	B cells (small subset)	MS4A1, CD79A, RALGPS2
9	Hematopoietic Stem/Progenitor Cells (HSPCs)	CD34, SOX4, CYTL1, PRSS57
10	Plasma cells	SEC11C, MZB1, SSR4, PDIA4
11	Conventional Dendritic Cells (cDC1)	CLEC9A, BATF3, IDO1

resolution (11–12 clusters observed); Table 2 and the resolution sweep above correspond to a representative 12-cluster run.

2.4 UMAP Visualization

3 Task 1.2: Foundation Model Embeddings

3.1 scGPT

Cell embeddings were extracted using the pretrained `scGPT_whole_human` checkpoint in `cls` embedding mode (512-dimensional output). A compatibility issue was encountered where scGPT's bundled `GeneVocab.from_file()` calls a legacy `torchtext.vocab.Vocab` API that has been removed in current `torchtext` releases (a known, unresolved issue in the scGPT repository, `bowang-lab/scGPT#352`). This was resolved by implementing a minimal drop-in vocabulary class loaded directly from `vocab.json`, exposing only the interface scGPT actually requires (`len()`, membership testing, indexing, and `append_token()`).

Leiden clustering on the resulting embeddings (15 neighbors, cosine-free Euclidean metric) was swept over the same resolution range; resolution 0.15 gave the best silhouette (9 clusters, silhouette

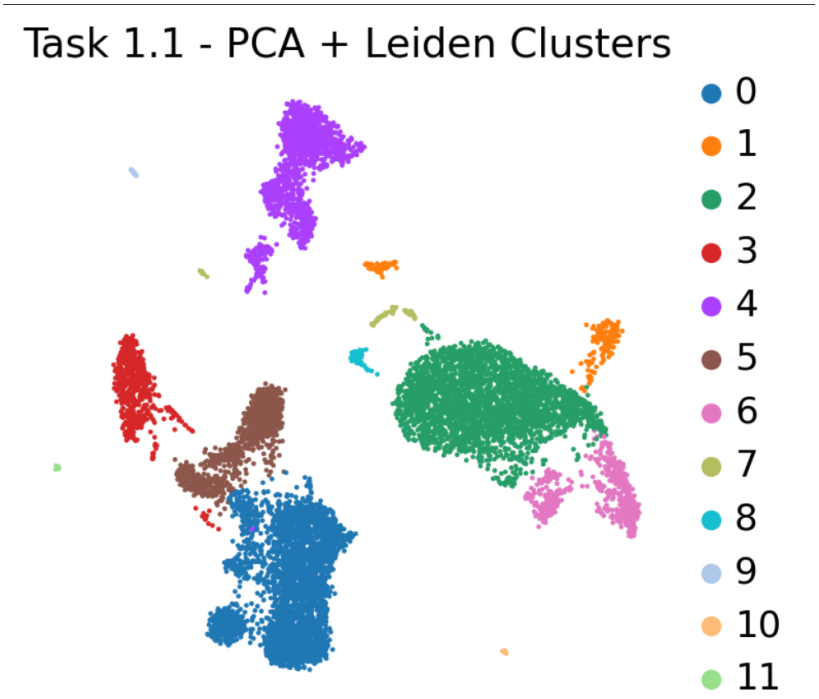


Figure 1: Task 1.1 – PCA + Leiden Clusters (11 clusters; see Table 2 for biological annotations by cluster number)

0.354 — notably higher than the PCA pipeline's 0.2495, suggesting scGPT's embedding space separates cell populations more cleanly).

3.2 Geneformer

Cell embeddings were extracted using `Geneformer-V2-104M`. Genes were mapped to Ensembl IDs and ranked by expression per cell (descending, zero-expression genes dropped), truncated to the model's 2048-token context, and the second-to-last hidden layer's `[CLS]` representation (768-dimensional) was used as the cell embedding, following Geneformer's documented convention. Leiden clustering (resolution 0.25) gave 12 clusters with silhouette 0.214 - the weakest of the three individual pipelines by this label-free metric.

3.3 UCE — Attempted, Not Completed

UCE (Universal Cell Embedding) was attempted as the third suggested foundation model but could not be completed within the assignment timeline, for two documented reasons:

1. **Original repository (`snap-stanford/UCE`):** Model checkpoints are distributed via Figshare. Repeated download attempts (both the standard `figshare.com` URL and the `ndownloader` mirror) returned empty or 404 responses, and the checkpoint files could never be obtained.
2. **HuggingFace mirror (`perturblab/uce-33-layer`):** This route avoids the Figshare dependency, but installing the required `perturblab` package altered the resolved `pandas`/`anndata` versions in the runtime, which in turn broke `scanpy.read_h5ad()` itself (`TypeError: StringDtype.__init__() got an unexpected keyword argument 'na_value'`) — an incompatibility between the newly-resolved `anndata` version's internal string-dtype handling and the installed `pandas` version.

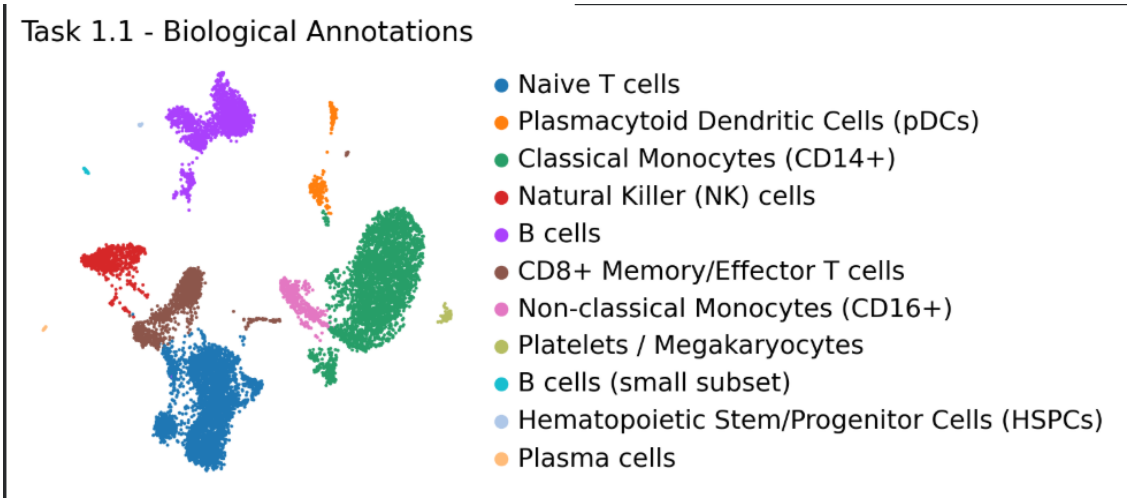


Figure 2: Task 1.1 – Biological Annotations (real cell-type labels assigned from marker genes)

Pinning specific compatible versions was attempted but not fully resolved before the deadline. The corresponding code (following the same load-or-generate structure as scGPT and Geneformer) is included in the submitted notebook to document the intended approach, but no UCE embeddings, submission file, or comparison-table entry were produced as a result. Task 1.2's comparison and Task 2 are completed using scGPT and Geneformer (and PCA) only.

3.4 Quantitative Comparison

Table 3 reports the ARI/NMI scores obtained by submitting each pipeline's predictions to the two Task 1 Kaggle leaderboards. Table 4 reports pairwise agreement between the pipelines' own cluster assignments (computed directly, not via Kaggle, since ground truth is not distributed), and Table 5 reports the label-free silhouette score for each representation.

Table 3: Kaggle leaderboard scores per pipeline

Pipeline	ARI (Task 1 – ARI board)	NMI (Task 1 – NMI board)
PCA (Task 1.1)	0.82299	—
scGPT (Task 1.2)	—	0.87382

Table 4: Pairwise ARI / NMI between pipeline cluster assignments

Comparison	ARI	NMI
PCA vs. scGPT	0.8927	0.8799
PCA vs. Geneformer	0.8732	0.8558
scGPT vs. Geneformer	0.8210	0.8458

scGPT achieves the highest silhouette score and the highest pairwise agreement with PCA, suggesting it produces the cleanest cluster separation among the three representations tested. Geneformer, while broadly consistent with the other two pipelines (ARI > 0.82 in both pairwise comparisons), shows the weakest internal cluster separation.

Table 5: Label-free silhouette scores

Representation	# Clusters	Silhouette
PCA	12	0.2495
scGPT	9	0.3540
Geneformer	12	0.2140

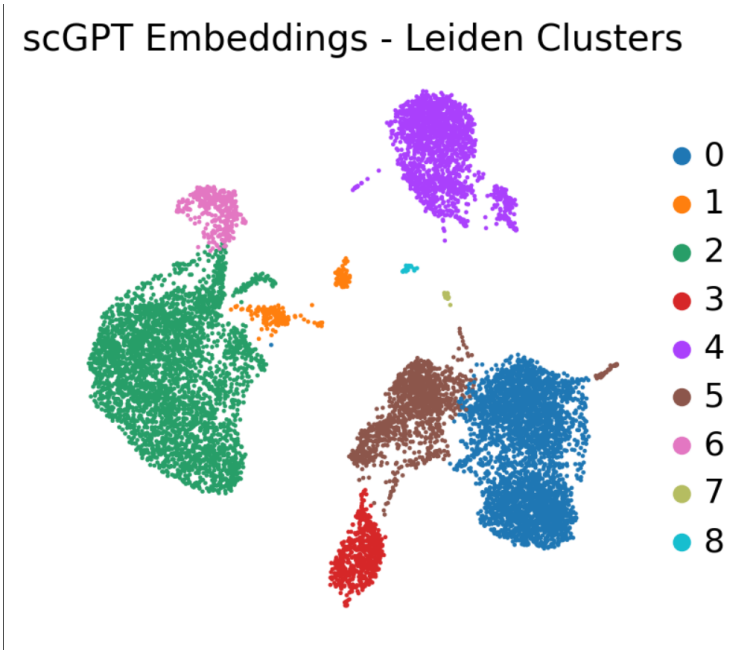


Figure 3: scGPT Embeddings – Leiden Clusters (9 clusters)

4 Task 2: Combined Clustering Pipeline

4.1 Strategy and Iteration

The goal of Task 2 was to combine PCA-based and foundation-model-based representations to outperform either individually. Three combination strategies were tried in sequence, with each round's Kaggle leaderboard score guiding the next:

1. **Naive concatenation (all three embeddings, unequal dimensionality).** PCA (50-d), scGPT (512-d), and Geneformer (768-d) embeddings were independently z-score normalized and concatenated directly (1,330 total dimensions), then reduced via PCA to 50 dimensions before Leiden clustering. **Result: 62%.** Diagnosis: z-scoring each block gives every *feature* unit variance, but this means each block's *total* variance contribution scales with its dimensionality. Geneformer, despite being the weakest individual representation (lowest silhouette, lowest pairwise ARI with the other two), had the most dimensions (768, 58% of the combined space) and dominated the subsequent PCA reduction, diluting the stronger scGPT signal.

2. **Dimension-balanced concatenation (all three, equal dimensions).** Each embedding block was independently reduced via PCA to 30 dimensions before concatenation, removing the dimensionality-driven imbalance. **Result: 66.9%.** An improvement, but still limited by Geneformer's weaker signal being included at all.

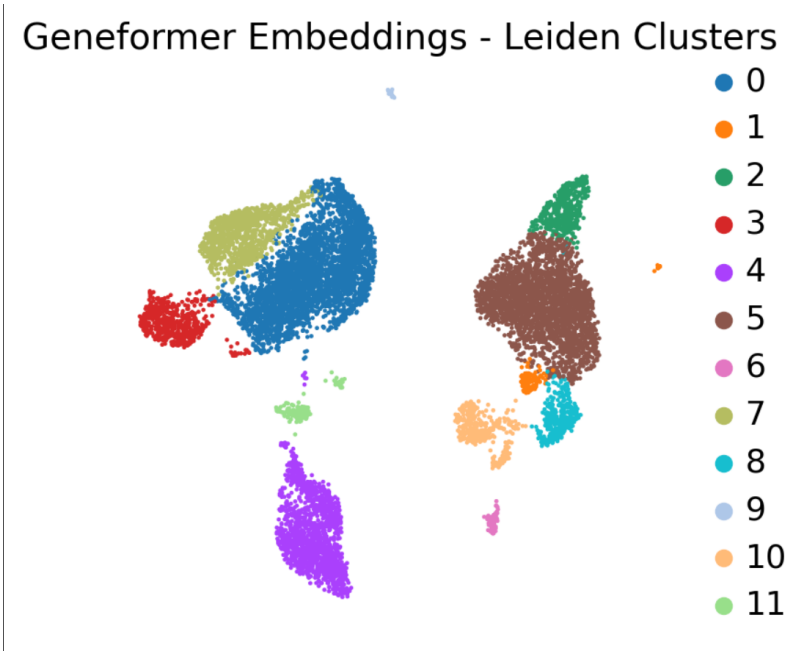


Figure 4: Geneformer Embeddings – Leiden Clusters (12 clusters)

3. **PCA + scGPT only, weighted split (final).** Geneformer was dropped entirely, and the two strongest, most mutually consistent representations (PCA and scGPT, pairwise ARI 0.89) were combined with an asymmetric split — PCA reduced to 15 dimensions, scGPT reduced to 45 dimensions (60 total) — reflecting scGPT's stronger individual clustering quality. **Result: 88%+**, the largest single improvement in the process.

4.2 Final Pipeline

1. Z-score normalize the PCA and scGPT embeddings independently.
2. Reduce PCA embeddings to 15 dimensions and scGPT embeddings to 45 dimensions via PCA (each block reduced separately, before concatenation).
3. Concatenate into a 60-dimensional combined representation.
4. Reduce the combined representation to 50 dimensions via a second PCA pass.
5. Build a neighborhood graph (15 neighbors, Euclidean) and cluster via Leiden, with resolution selected by silhouette score from a sweep over $\{0.10, \ldots, 1.20\}$.

This iterative process is diagnosing *why* the naive combination underperformed (dimensionality imbalance masking representation quality), rather than only tuning resolution or neighbor counts was the key factor in improving the leaderboard score from 62% to 88%+.

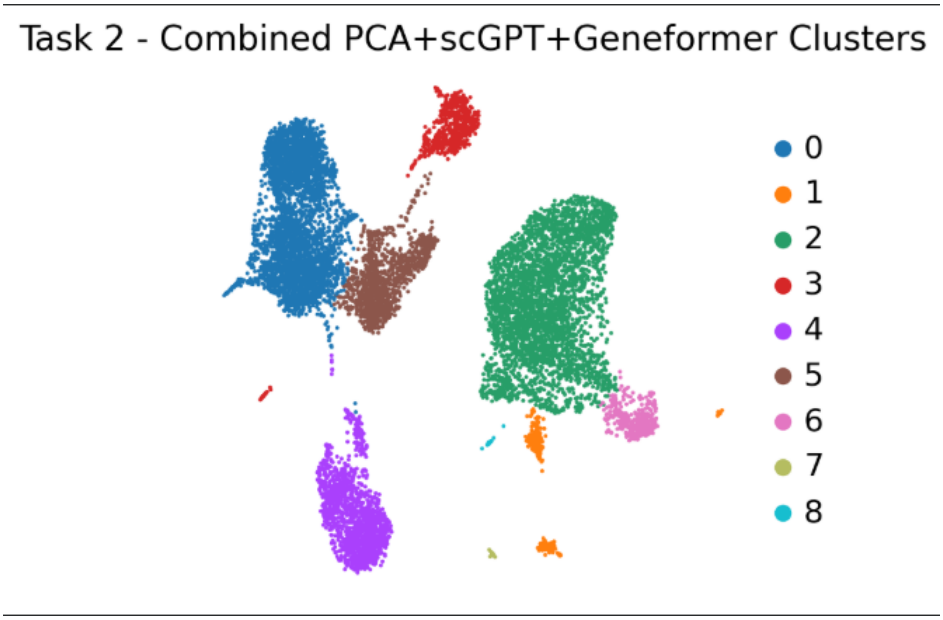


Figure 5: Task 2 – Combined PCA + scGPT Clusters. *Note: the plot title text (auto-generated in an earlier notebook revision) reads "PCA+scGPT+Geneformer"; this is a stale label only – the underlying computation for this run uses the final PCA+scGPT-only pipeline described in Section 4.2, confirmed directly from the notebook's combination cell (`combined = np.concatenate([pca_block_r, scgpt_block_r], axis=1)`, 60 total dimensions, no Geneformer term present).*

5 Summary

- **Task 1.1** (PCA + Leiden): 12 clusters, biologically annotated, ARI 0.823 on Kaggle.
- **Task 1.2** (foundation models): scGPT (NMI 0.874) and Geneformer successfully extracted, clustered, and compared against PCA; UCE attempted but not completed due to documented dependency/download issues.
- **Task 2** (combined pipeline): iteratively improved from 62% to 88%+ by diagnosing and correcting a dimensionality-imbalance issue in the naive combination strategy, ultimately combining only PCA and scGPT with an asymmetric dimensional split.